1. Write a program for creating the new activity for the button click.

MainActivity.java

```
package com.example.myapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        btn.setOnClickListener(v -> {
            Intent i = new Intent(MainActivity.this,
                    SecondActivity.class);
            startActivity(i);
        });
    }
}
```

SecondActivity.java

```
package com.example.myapp;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

---

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<Button
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Open Activity"/>
```

---

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
    android:text="Second Activity"/>
```

```
AndroidManifest.xml
<activity android:name=".SecondActivity"/>
```

2. Write a program for creating the two fragments on single screen.

```
MainActivity.java
package com.example.myapp;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```
activity_main.xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <fragment
        android:name="com.example.myapp.FirstFragment"
        android:layout_width="match_parent"
        android:layout_height="100dp"/>

    <fragment
        android:name="com.example.myapp.SecondFragment"
        android:layout_width="match_parent"
        android:layout_height="100dp"/>
</LinearLayout>
```

```
FirstFragment.java
package com.example.myapp;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;

public class FirstFragment extends Fragment {

    @Override
```

```
    public View onCreateView(LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {

        return inflater.inflate(
                R.layout.fragment_first,
                container,
                false);
    }
}
```

```
SecondFragment.java
package com.example.myapp;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;

public class SecondFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {

        return inflater.inflate(
                R.layout.fragment_second,
                container,
                false);
    }
}
```

```
fragment_first.xml
<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="First Fragment"
    android:gravity="center"/>
```

```
fragment_second.xml
<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="Second Fragment"
    android:gravity="center"/>
```

3. Write an application for applying the linear layout.

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Linear Layout"/>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"/>
</LinearLayout>
```

4. Write an application for applying the relative layout.

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/text1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello"/>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"
        android:layout_below="@id/text1"/>
</RelativeLayout>
```

5. Write an application for applying the absolute layout.

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">
```

```
    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"
        android:layout_x="100dp"
        android:layout_y="100dp"/>
</AbsoluteLayout>
```

6. Write an application for applying the ListView layout.

MainActivity.java

```java
package com.example.myapp;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ListView listView;

    String[] items = {"Apple", "Banana", "Mango", "Orange"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);

        ArrayAdapter<String> adapter =
                new ArrayAdapter<>(this,
                        android.R.layout.simple_list_item_1,
                        items);

        listView.setAdapter(adapter);
    }
}
```

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<ListView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/listView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

7. Write an application for applying the Table layout.

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<TableLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TableRow>

        <TextView
            android:text="Name"
            android:padding="10dp"/>

        <TextView
            android:text="Age"
            android:padding="10dp"/>
    </TableRow>

    <TableRow>

        <TextView
            android:text="Rohit"
            android:padding="10dp"/>

        <TextView
            android:text="21"
            android:padding="10dp"/>
    </TableRow>

</TableLayout>
```

8. Write an application for applying the Grid layout.

```xml
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<GridLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:columnCount="2"
    android:rowCount="2">

    <Button
        android:text="1"/>

    <Button
        android:text="2"/>

    <Button
        android:text="3"/>

    <Button
        android:text="4"/>
</GridLayout>
```

9. Write an application for applying the Frame layout.

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<FrameLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Frame Layout"
        android:textSize="25sp"/>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Button"/>
</FrameLayout>
```

10. Write a program for implicit intent.

```
MainActivity.java

package com.example.myapp;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        btn.setOnClickListener(v -> {

            Intent intent = new Intent(Intent.ACTION_VIEW);

intent.setData(Uri.parse("https://www.google.com"));

            startActivity(intent);
        });
```

```
    }
}
```

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Google"/>
</LinearLayout>
```

11. Write a program for explicit intent.

MainActivity.java

```java
package com.example.myapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        btn.setOnClickListener(v -> {

            Intent intent =
                    new Intent(MainActivity.this,
                            SecondActivity.class);

            startActivity(intent);
        });
    }
}
```

SecondActivity.java

```java
package com.example.myapp;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

---

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Open Activity"/>
</LinearLayout>
```

---

activity_second.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Second Activity"/>
</LinearLayout>
```

Add in AndroidManifest.xml

```xml
<activity android:name=".SecondActivity"/>
```

12. Write a program for broadcast receiver.

```
MainActivity.java

package com.example.myapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        btn.setOnClickListener(v -> {

            Intent intent = new Intent("MY_MESSAGE");
            sendBroadcast(intent);

        });
    }
}
```

```
MyReceiver.java

package com.example.myapp;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.Toast;

public class MyReceiver extends BroadcastReceiver {

    @Override
    public void onReceive(Context context, Intent intent) {

        Toast.makeText(context,
                "Broadcast Received",
                Toast.LENGTH_LONG).show();
    }
}
```

```
activity_main.xml
```

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Broadcast"/>
</LinearLayout>
```

```
Add in AndroidManifest.xml

<receiver android:name=".MyReceiver">
    <intent-filter>
        <action android:name="MY_MESSAGE"/>
    </intent-filter>
</receiver>
```

13. Write a program for designing the registration form GUI using Buttons, Text Fields, Checkboxes, Radio Buttons, and Toggle Buttons with their event handler.

14. Write a program for Spinner using event handler to display selected option name.

```java
MainActivity.java

package com.example.myapp;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Spinner spinner;

    String[] items = {"Java", "Python", "Android"};

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        spinner = findViewById(R.id.spinner);
```

```
        ArrayAdapter<String> adapter =
                new ArrayAdapter<>(this,
                        android.R.layout.simple_spinner_item,
                        items);

        adapter.setDropDownViewResource(
android.R.layout.simple_spinner_dropdown_item);

        spinner.setAdapter(adapter);

        spinner.setOnItemSelectedListener(
                new AdapterView.OnItemSelectedListener() {

            @Override
            public void onItemSelected(AdapterView<?> parent,
                                       View view,
                                       int position,
                                       long id) {

                Toast.makeText(MainActivity.this,
                        items[position],
                        Toast.LENGTH_SHORT).show();
            }

            @Override
            public void onNothingSelected(AdapterView<?>
parent) {

            }
        });
    }
}
```

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Spinner
        android:id="@+id/spinner"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>
</LinearLayout>
```

15. Write a program for Alert Dialog using event handler.

```
MainActivity.java
```

```
package com.example.myapp;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        btn.setOnClickListener(v -> {

            AlertDialog.Builder builder =
                    new
AlertDialog.Builder(MainActivity.this);

            builder.setTitle("Alert");
            builder.setMessage("Do you want to continue?");

            builder.setPositiveButton("Yes",
                    (dialog, which) ->
                            Toast.makeText(MainActivity.this,
                                    "Yes Clicked",

Toast.LENGTH_SHORT).show());

            builder.setNegativeButton("No",
                    (dialog, which) ->
                            Toast.makeText(MainActivity.this,
                                    "No Clicked",

Toast.LENGTH_SHORT).show());

            builder.show();
        });
    }
}
```

---

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
```

```
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Alert"/>
</LinearLayout>
```

16. Write a program for demonstrating Option Menu along with event handler.

```
MainActivity.java

package com.example.myapp;

import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {

        menu.add("Home");
        menu.add("About");

        return true;
    }

    @Override
    public boolean onOptionsItemSelected(MenuItem item) {

        Toast.makeText(this,
                item.getTitle(),
                Toast.LENGTH_SHORT).show();

        return true;
    }
}
```

```
activity_main.xml
```

```xml
<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="Option Menu"
    android:gravity="center"/>
```

17. Write a program for demonstrating Context Menu along with event handler.

```java
MainActivity.java

package com.example.myapp;

import android.os.Bundle;
import android.view.ContextMenu;
import android.view.MenuItem;
import android.view.View;
import android.widget.TextView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView text;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        text = findViewById(R.id.text);

        registerForContextMenu(text);
    }

    @Override
    public void onCreateContextMenu(ContextMenu menu,
                                View v,

ContextMenu.ContextMenuInfo menuInfo) {

        menu.add("Edit");
        menu.add("Delete");
    }

    @Override
    public boolean onContextItemSelected(MenuItem item) {

        Toast.makeText(this,
                item.getTitle(),
                Toast.LENGTH_SHORT).show();

        return true;
```

```
    }
}
```

---

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/text"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="Long Press Here"
    android:gravity="center"/>
```

18. Write a program for demonstrating Popup Menu along with event handler.

MainActivity.java

```
package com.example.myapp;

import android.os.Bundle;
import android.widget.Button;
import android.widget.PopupMenu;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        btn.setOnClickListener(v -> {

            PopupMenu popupMenu =
                    new PopupMenu(MainActivity.this, btn);

            popupMenu.getMenu().add("Save");
            popupMenu.getMenu().add("Exit");

            popupMenu.setOnMenuItemClickListener(item -> {

                Toast.makeText(MainActivity.this,
                        item.getTitle(),
                        Toast.LENGTH_SHORT).show();

                return true;
            });
```

```
            popupMenu.show();
        });
    }
}
```

---

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Popup"/>
</LinearLayout>
```

19. Write a program for demonstrating Notification.

MainActivity.java

```
package com.example.myapp;

import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;

public class MainActivity extends AppCompatActivity {

    String CHANNEL_ID = "MyChannel";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

            NotificationChannel channel =
                    new NotificationChannel(
```

```
                            CHANNEL_ID,
                            "MyChannel",

NotificationManager.IMPORTANCE_DEFAULT);

            NotificationManager manager =

getSystemService(NotificationManager.class);

            manager.createNotificationChannel(channel);
        }

        btn.setOnClickListener(v -> {

            NotificationCompat.Builder builder =
                    new NotificationCompat.Builder(
                            MainActivity.this,
                            CHANNEL_ID)

                    .setContentTitle("Notification")
                    .setContentText("Hello Android")

.setSmallIcon(android.R.drawable.ic_dialog_info);

            NotificationManager manager =
                    (NotificationManager)

getSystemService(NOTIFICATION_SERVICE);

            manager.notify(1, builder.build());
        });
    }
}
```

`activity_main.xml`

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Notification"/>
</LinearLayout>
```

20. Write a program for inserting data into database using SQLite.

`MainActivity.java`

```java
package com.example.myapp;

import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    SQLiteDatabase db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        db = openOrCreateDatabase(
                "StudentDB",
                MODE_PRIVATE,
                null);

        db.execSQL("CREATE TABLE IF NOT EXISTS Student(name
VARCHAR)");

        btn.setOnClickListener(v -> {

            db.execSQL("INSERT INTO Student VALUES('Rohit')");

            Toast.makeText(this,
                    "Data Inserted",
                    Toast.LENGTH_SHORT).show();
        });
    }
}
```

---

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Insert Data"/>
</LinearLayout>
```

21. Write a program for displaying data from database using SQLite.

```
MainActivity.java

package com.example.myapp;

import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    SQLiteDatabase db;
    TextView text;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        text = findViewById(R.id.text);

        db = openOrCreateDatabase(
                "StudentDB",
                MODE_PRIVATE,
                null);

        db.execSQL("CREATE TABLE IF NOT EXISTS Student(name
VARCHAR)");

        db.execSQL("INSERT INTO Student VALUES('Rohit')");

        Cursor c = db.rawQuery(
                "SELECT * FROM Student",
                null);

        c.moveToFirst();

        text.setText(c.getString(0));
    }
}
```

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<TextView
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/text"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:textSize="25sp"/>
```

22. Write a program for updating the data of database using SQLite.

```
MainActivity.java

package com.example.myapp;

import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    SQLiteDatabase db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        db = openOrCreateDatabase(
                "StudentDB",
                MODE_PRIVATE,
                null);

        db.execSQL("CREATE TABLE IF NOT EXISTS Student(name
VARCHAR)");

        db.execSQL("INSERT INTO Student VALUES('Rohit')");

        btn.setOnClickListener(v -> {

            db.execSQL(
                    "UPDATE Student SET name='Ram' WHERE
name='Rohit'");

            Toast.makeText(this,
                    "Data Updated",
                    Toast.LENGTH_SHORT).show();
        });
    }
}
```

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Update Data"/>
</LinearLayout>
```

23. Write a program for storing the data into file.

```
MainActivity.java

package com.example.myapp;

import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.io.FileOutputStream;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        Button btn = findViewById(R.id.button);

        btn.setOnClickListener(v -> {

            String data = "Hello Android";

            try {

                FileOutputStream fos =
                        openFileOutput(
                                "myfile.txt",
                                MODE_PRIVATE);

                fos.write(data.getBytes());
                fos.close();

                Toast.makeText(this,
                        "File Saved",
```

```
                    Toast.LENGTH_SHORT).show();
            } catch (Exception e) {
                e.printStackTrace();
            }
        });
    }
}
```

---

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/button"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Save File"/>
</LinearLayout>
```

24. Write an application for multithreading.

MainActivity.java

```
package com.example.myapp;

import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView text;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        text = findViewById(R.id.text);

        Thread thread = new Thread(() -> {

            try {
```

```
                for (int i = 1; i <= 5; i++) {

                    int num = i;

                    runOnUiThread(() ->
                            text.setText("Count : " + num));

                    Thread.sleep(1000);
                }

            } catch (Exception e) {

                e.printStackTrace();
            }
        });

        thread.start();
    }
}
```

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/text"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:textSize="30sp"/>
```

25. Write a program for demonstrating Progress Bar along with the progress percentage to be displayed on screen.

```
MainActivity.java

package com.example.myapp;

import android.os.Bundle;
import android.os.Handler;
import android.widget.ProgressBar;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ProgressBar progressBar;
    TextView text;
    int progress = 0;

    @Override
```

```java
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        progressBar = findViewById(R.id.progressBar);
        text = findViewById(R.id.text);

        Handler handler = new Handler();

        new Thread(() -> {

            while (progress <= 100) {

                int value = progress;

                handler.post(() -> {

                    progressBar.setProgress(value);
                    text.setText(value + "%");
                });

                progress++;

                try {
                    Thread.sleep(100);
                } catch (Exception e) {
                    e.printStackTrace();
                }
            }

        }).start();
    }
}
```

---

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <ProgressBar
        android:id="@+id/progressBar"
        style="?android:attr/progressBarStyleHorizontal"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:max="100"/>

    <TextView
        android:id="@+id/text"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="0%"
        android:textSize="20sp"/>
</LinearLayout>
```

26. Write a program for demonstrating Rating Bar along with display of the rating value selected.

```
MainActivity.java

package com.example.myapp;

import android.os.Bundle;
import android.widget.RatingBar;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    RatingBar ratingBar;
    TextView text;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        ratingBar = findViewById(R.id.ratingBar);
        text = findViewById(R.id.text);

        ratingBar.setOnRatingBarChangeListener(
                (ratingBar, rating, fromUser) -> {

            text.setText("Rating : " + rating);
        });
    }
}
```

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <RatingBar
        android:id="@+id/ratingBar"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Rating : 0"
        android:textSize="20sp"/>
</LinearLayout>
```

27. Write a program for demonstrating Seek Bar along with event handler.

```
MainActivity.java

package com.example.myapp;

import android.os.Bundle;
import android.widget.SeekBar;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    SeekBar seekBar;
    TextView text;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        seekBar = findViewById(R.id.seekBar);
        text = findViewById(R.id.text);

        seekBar.setOnSeekBarChangeListener(
                new SeekBar.OnSeekBarChangeListener() {

            @Override
            public void onProgressChanged(SeekBar seekBar,
                                          int progress,
                                          boolean fromUser) {

                text.setText("Value : " + progress);
            }

            @Override
            public void onStartTrackingTouch(SeekBar seekBar)
{

            }

            @Override
            public void onStopTrackingTouch(SeekBar seekBar) {
```

```
            }
        });
    }
}
```

```
activity_main.xml

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <SeekBar
        android:id="@+id/seekBar"
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:max="100"/>

    <TextView
        android:id="@+id/text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Value : 0"
        android:textSize="20sp"/>
</LinearLayout>
```

28. Write a program for demonstrating Toggle Button along with event handler.

```
MainActivity.java

package com.example.myapp;

import android.os.Bundle;
import android.widget.TextView;
import android.widget.ToggleButton;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ToggleButton toggleButton;
    TextView text;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
```

```java
        toggleButton = findViewById(R.id.toggleButton);
        text = findViewById(R.id.text);

        toggleButton.setOnCheckedChangeListener(
                (buttonView, isChecked) -> {

            if (isChecked) {
                text.setText("Toggle ON");
            } else {
                text.setText("Toggle OFF");
            }
        });
    }
}
```

---

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"/>

    <TextView
        android:id="@+id/text"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Toggle OFF"
        android:textSize="20sp"/>
</LinearLayout>
```

29. Write a program for Calculator.

MainActivity.java

```java
package com.example.myapp;

import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
```

```java
    EditText n1, n2;
    Button add;
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        n1 = findViewById(R.id.n1);
        n2 = findViewById(R.id.n2);
        add = findViewById(R.id.button);
        result = findViewById(R.id.result);

        add.setOnClickListener(v -> {

            int a = Integer.parseInt(n1.getText().toString());
            int b = Integer.parseInt(n2.getText().toString());

            int c = a + b;

            result.setText("Result : " + c);
        });
    }
}
```

---

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/n1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter First Number"
        android:inputType="number"/>

    <EditText
        android:id="@+id/n2"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Second Number"
        android:inputType="number"/>

    <Button
        android:id="@+id/button"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Add"/>

    <TextView
        android:id="@+id/result"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Result"
        android:textSize="20sp"/>
</LinearLayout>
```